

CSC1005/CSC1022 Database Lab

Lab 6: SQL Joins and ANY/ALL

Goal: Practice miscellaneous SQL queries and try out EXISTS, LIMIT clauses.

You have finished the lab when you have finished all queries.

Section 1: Basketball (Make default database)

1. Return the max height among the players playing with 'Chicago Bulls' in the year 1998.

Ans: Max(height)
84

```
SELECT max(height)
FROM teams, players_teams, players
where teams.name = 'Chicago Bulls' and teams.tmID = players_teams.tmID and
players_teams.playerID = players.playerID and players_teams.year = 1998
```

2. Return the average height for each team in the year 1998, ordered from the highest to the lowest.

Ans:

```
SELECT teams.name, avg(height)
FROM teams, players_teams, players
where teams.tmID = players_teams.tmID and players_teams.playerID = players.playerID and
players_teams.year = 1998
group by teams.name
order by avg(height) desc
```

Section 2: Sakila (Make Sakila the default database)

3. Find all actors who have appeared in at least one film. (Use **EXISTS** keyword)

Ans:

```
SELECT actor_id, first_name, last_name
FROM actor a
WHERE EXISTS (
    SELECT fa.film_id
    FROM film_actor fa
    WHERE fa.actor_id = a.actor_id
);
```

4. Count the number of movies belonging to the 'Action' category

Ans: 64

```
SELECT count(*)  
FROM film_category, category  
where category.category_id=film_category.category_id and  
category.name = 'Action'
```

Section 3. World (Make default database)

5. Return a table with the name of the country and the average population of its cities.

Ans:

```
SELECT Country.Name, avg(City.Population)  
FROM City, Country  
WHERE City.CountryCode = Country.Code  
group by Country.Name
```

6. List the top 10 most used languages, the number of countries which speak the language and total amount of people who speak this language (not the percent). Order the results by the total number of people who speak these languages from highest to lowest.

Ans:

```
SELECT CL.Language, COUNT(CL.Language) AS CNbr, SUM(C.Population*CL.Percentage) AS  
Total  
FROM country C,countrylanguage CL  
WHERE C.Code=CL.CountryCode  
GROUP BY CL.Language  
ORDER BY Total DESC  
LIMIT 20;
```